

Requirement Specification Document

Project: Women and Child Development Department Portal

Client: Women and Child Development Department, Government of Sikkim

1. Introduction

The Women and Child Development Department, Government of Sikkim, is developing an online portal to manage and administer welfare schemes for citizens across various categories such as age, gender, caste, and other eligibility parameters.

This document defines the biometric authentication and UIDAI integration requirements to be implemented by the biometric device vendor.

2. Scope of Work

The biometric vendor is required to:

- Provide UIDAI-certified biometric devices.
- Integrate Aadhaar authentication APIs.
- Ensure secure communication with UIDAI servers.
- Return structured authentication responses to the portal backend.

3. Biometric Authentication Instances

Biometric authentication is mandatory in the following scenarios:

Instance 1: Registration Verification

- When a newly registered user visits the office for document verification.
- Biometric authentication confirms identity and activates verified status.

Instance 2: Scheme Application Verification

- Every time a verified user applies for a scheme.
- Biometric authentication is required before the application moves to the next approval level.

Note: If a user applies for multiple schemes, biometric authentication must be performed separately for each application.

4. Functional Requirements

FR-01: Biometric verification during initial registration verification.
FR-02: Biometric verification for every scheme application.
FR-03: Real-time Aadhaar authentication via UIDAI API.
FR-04: Secure encrypted communication (HTTPS, TLS 1.2+).
FR-05: Transaction logging and audit trail maintenance.

5. API Requirements

The biometric system must:

- Capture fingerprint using UIDAI-certified device.
- Encrypt biometric data as per UIDAI standards.
- Send authentication request to UIDAI server.
- Return structured response containing authentication status and transaction ID.

Sample Response Structure:

```
{
  "status": "SUCCESS/FAILURE",
  "aadhaarMasked": "XXXX-XXXX-1234",
  "name": "Applicant Name",
  "dob": "YYYY-MM-DD",
  "gender": "M/F/O",
  "transactionId": "UIDAI_TXN_ID",
  "phoneNo": "XXXXXX1234",
  "address": [ ]
}
```

We would require the following from Aadhaar Database for a particular User - Aadhaar Number (masked), Phone number linked with Aadhaar (masked), Full Name, DOB (DD-MM-YYYY or YYYY-MM-DD), Gender, Address Details and other relevant information related to the API call like status, transactionID, etc.

6. Security & Compliance

- No storage of raw biometric data.
- Masked Aadhaar number storage only.
- Complete audit trail for each transaction.
- Device ID and Officer ID logging.
- Compliance with UIDAI authentication guidelines.

7. Audit Log Requirements

Each biometric transaction must record:

- Registration/Application ID

- Masked Aadhaar Number
- Timestamp
- Officer ID
- Device IDA
- Authentication Result
- UIDAI Transaction ID

8. Non-Functional Requirements

- Authentication response time ≤ 5 seconds.
- High availability during office hours.
- Scalable across multiple offices in Sikkim.
- Secure deployment in government network infrastructure.

9. Conclusion

Biometric authentication is a mandatory security control to prevent impersonation and ensure authentic Aadhaar-based identity verification.

It shall be implemented:

- Once during user registration verification.
- Once for every individual scheme application.

This ensures transparency, accountability, and secure welfare distribution across the state.